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#include <iostream>
#include <cstdlib>

using namespace std;

template<class T>

class Memory
{
public:
static T** allocArray(int m, int n)
{
// Allocate memory for an array of pointers (each pointing to a row)
T** array = (T**) malloc(m * sizeof(T*));

// Allocate a single contiguous block of memory for m * n elements
T* data = (T*) malloc(m * n * sizeof(T));

// Set each row pointer to point to the correct part of the contiguous memory block
for (int i = 0; i < m; i++)
{
array[i] = &data[i * n];
}

return array;
}
};

int main()
{
int **array;
array = Memory<int>::allocArray(5, 10);
int j, k;

for (j = 0; j < 5; j++)
{
for (k = 0; k < 10; k++)
{
array[j][k] = j * 10 + k;
}
}
for (j = 0; j < 5; j++)
{
for (k = 0; k < 10; k++)
{
cout << array[j][k] << " ";
}
}

return 0;
}

```